

西安交通大学本科生课程考试试题标准答案与评分标准

课程名称: 工程力学 A 课时: 56 考试时间: 2025 年 月 日

1. 填空题 (本大题共 10 小题, 每小题 3 分, 共 30 分)

(1) 60° (或 $\frac{\pi}{3}$) (3 分)

(2) $(0.6F)$ (1 分), $(-2.4Fa)$ (1 分), $\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4a & 4a & 0 \\ -0.4\sqrt{2}F & -0.4\sqrt{2}F & 0.6F \end{vmatrix}$ (1 分)

(3) $F(\mathbf{i}+\mathbf{k})$ (1 分), $Fa(\mathbf{i}-\mathbf{k})$ (1 分), 合力 (1 分)

(4) $\frac{\sqrt{3}}{2}l\omega$ (2 分), 水平向左 (1 分)

(5) $2a$ (2 分), 150° (或 $\frac{5\pi}{6}$) (1 分)

(6) $\frac{\pi}{4}d^2$ (1 分), $d\delta$ (1 分), $\frac{2F}{\pi d^2} \leq [\tau]$ (1 分) (7) 30 (2 分), AB (1 分)

(8) $\frac{8EI}{ql^2}$ (3 分) (9) $\frac{Fl^3}{4EI}$ (3 分) (10) $\frac{\pi d^4}{64}$ (1 分), $\frac{\pi^3 Ed^4}{256l^2}$ (2 分)

2. 计算题 (10 分)

解: (1) CD 为二力杆, 杆 BCE 受力如图 (a) (2 分)

$$\sum M_B(F) = 0 \quad F_{CD} \cdot a - M = 0 \quad F_{CD} = F \quad (1 \text{ 分})$$

$$\sum F_x = 0 \quad F_{CD} - F_{Bx} = 0 \quad F_{Bx} = F \quad (1 \text{ 分})$$

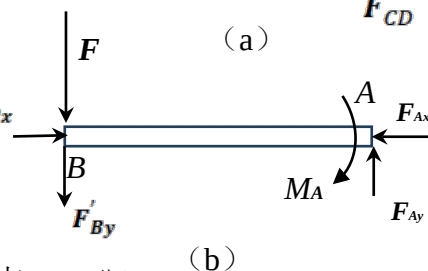
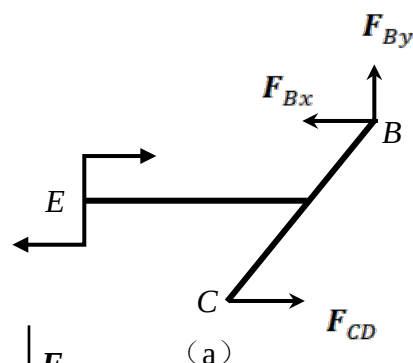
$$\sum F_y = 0 \quad F_{By} = 0 \quad (2 \text{ 分}) \quad (1 \text{ 分})$$

(2) 杆 AB 受力如图 (b) (2 分)

$$\sum F_x = 0 \quad F'_{Bx} - F_{Ax} = 0 \quad F_{Ax} = F \quad (1 \text{ 分})$$

$$\sum F_y = 0 \quad F_{Ay} - F - F'_{By} = 0 \quad F_{Ay} = F \quad (1 \text{ 分})$$

$$\sum M_A(F) = 0 \quad F \cdot a - M_A = 0 \quad M_A = Fa \quad (\text{顺时针}) \quad (1 \text{ 分})$$



3.计算题 (10 分)

解: 此为一次超静定问题, 杆受力如图所示

(1) 平衡方程:

$$\sum F_x = 0: F_A = F_B \quad (1 \text{ 分})$$

(2) 变形协调方程

$$\Delta L = \Delta l_{AC} + \Delta l_{CD} + \Delta l_{DB} = 0 \quad (1 \text{ 分})$$

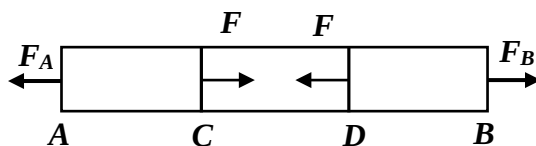
(3) 物理方程

$$\Delta l_{AC} = \frac{F_A l}{EA}, \quad \Delta l_{CD} = \frac{(F_A - F) l}{EA}, \quad \Delta l_{DB} = \frac{F_B l}{EA} \quad (2 \text{ 分})$$

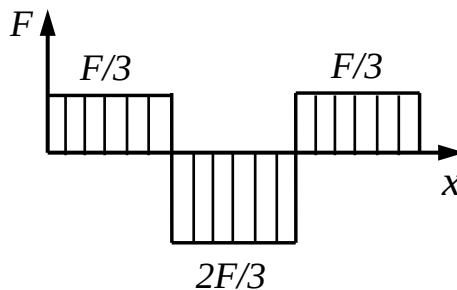
$$\text{补充方程: } 2F_A + F_B = F \quad (1 \text{ 分})$$

$$\text{则: } F_A = F_B = \frac{F}{3}, \quad F_{CD} = -\frac{2F}{3} \quad (1 \text{ 分})$$

$$\text{最大应力为 } \sigma_{\max} = \left| \frac{F_{CD}}{A} \right| = \frac{2F}{3A} \quad (1 \text{ 分})$$



(1 分)

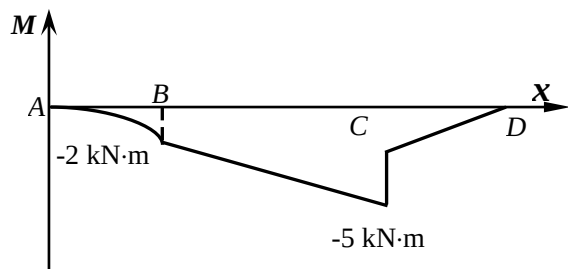
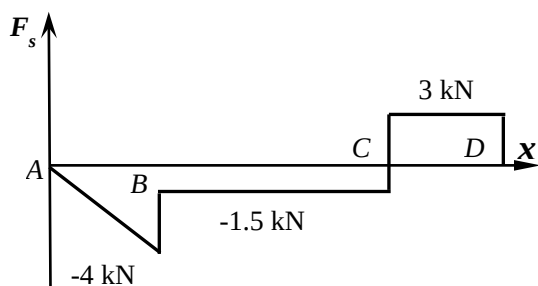


(2 分)

4.计算题 (10 分)

$R_B = 2.5 \text{ kN}$ (向上), $R_C = 4.5 \text{ kN}$ (向上) (2 分)

$|F_s|_{\max} = 4 \text{ kN}$, $|M|_{\max} = 5 \text{ kN}\cdot\text{m}$ (2 分)



(每个图各 3 分)

5. 计算题 (12 分)

解: 取套筒 A 为动点, 动系固连于 BC 杆上, 牵连运动为平动 [1 分]

$$\vec{v}_A = \vec{v}_A^e + \vec{v}_A^r$$

$$v_A = \omega L = 30 \text{ cm/s}, \quad v_A^e = v_A \tan \varphi, \quad v_A^e = \omega L \tan \varphi = 30\sqrt{3} \text{ cm/s}$$

$$\omega_1 = \frac{v_A^e}{O_1 B} = 1.5 \text{ rad/s} \quad (\text{转向如图})$$

[图 2 分+计算 2 分]

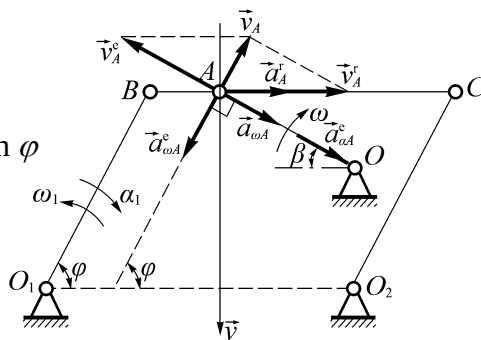
$$\vec{a}_{\omega A} = \vec{a}_{\alpha A}^e + \vec{a}_{\omega A}^e + \vec{a}_A^r$$

向 Ay 轴投影, $a_{\omega A} \cos \varphi = a_{\alpha A}^e \cos \varphi + a_{\omega A}^e \sin \varphi$

$$a_{\omega A} = \omega^2 L = 60 \text{ cm/s}^2$$

$$a_{\alpha A}^e = O_1 B \omega_1^2 = 45\sqrt{3} \text{ cm/s}^2$$

$$\alpha_1 = \frac{a_{\alpha A}^e}{O_1 B} = -\frac{75}{20\sqrt{3}} = -1.25\sqrt{3} = -2.16 \text{ rad/s}^2 \quad (\text{逆转向}) \quad [\text{图 2 分+计算 5 分}]$$



6 计算题 (15 分)

(1) 解: 取 A 点为基点, B 点为动点

OA 杆作定轴转动, $v_A = 2\omega_0 l$; $v_A \perp OA$

O1B 杆作定轴转动, $v_B \perp O1B$

$$v_B = v_A + v_{BA}; \quad v_{BA} \perp AB$$

解得: $v_B = v_A = v_{BA} = 2\omega_0 l$

$$\text{因此 } \omega_{O1B} = \frac{v_B}{BO_1} = 2\omega_0 \quad (5 \text{ 分})$$

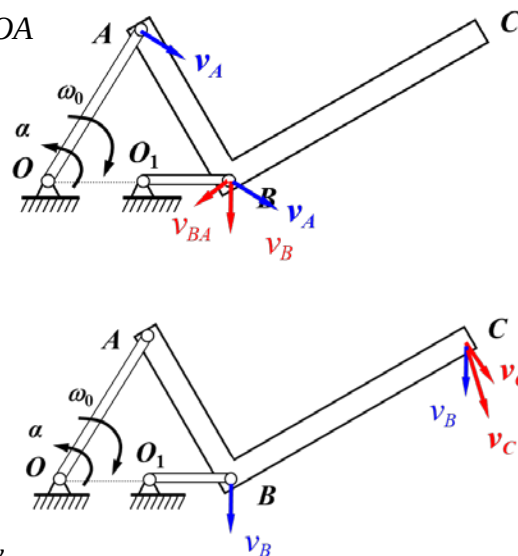
(2) 以 B 点为基点, C 点为动点

由 (1) 得到 $v_{BA} = 2\omega_0 l$, 因此 $\omega_{ABC} = \frac{v_{BA}}{AB} = \omega_0$

$$v_C = v_B + v_{CB}; \quad v_{CB} \perp CB$$

$$v_{CB} = \omega_{ABC} \cdot BC = 2\sqrt{3}\omega_0 l$$

由几何关系解得: $v_C = \sqrt{v_B^2 + v_{CB}^2 - 2v_B v_{CB} \cos 150^\circ} = 2\sqrt{7}\omega_0 l \quad (5 \text{ 分})$

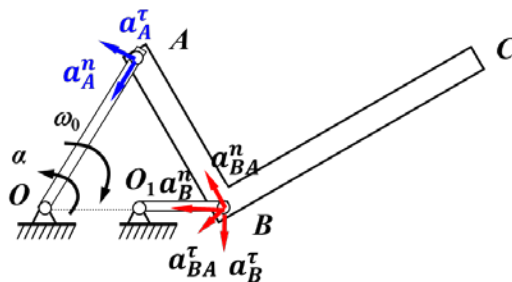


(3) 以 A 点为基点, B 点为动点

$$a_B^n + a_B^r = a_A^n + a_A^r + a_{BA}^n + a_{BA}^r$$

大小: $\checkmark$? $\checkmark$ $\checkmark$ $\checkmark$?

方向: $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$ $\checkmark$



其中: $a_B^n = 4\omega_0^2 l$; $a_A^n = 2\omega_0^2 l$; $a_A^r = 2\alpha l$; $a_{BA}^n = \omega_{ABC}^2 l_{AB} = 2\omega_0^2 l$

沿 AB 杆方向投影: $a_B^r \sin 60^\circ - a_B^n \cos 60^\circ = a_A^n \cos 60^\circ - a_A^r \sin 60^\circ - a_{BA}^n$

解得 $a_B^r = \frac{2\sqrt{3}}{3} \omega_0^2 l - 2\alpha l$;

因此, B 点的加速度为: $a_B^r = \frac{2\sqrt{3}}{3} \omega_0^2 l - 2\alpha l$; $a_B^n = 4\omega_0^2 l$ (5 分)

7. 计算题 (13 分)

解: $F_N = 40 \text{ kN}$, $\sigma_{F_N} = \frac{F_N}{A} = 8.0 \text{ MPa}$ 。(2 分)

$T = m = 2 \text{ kN} \cdot \text{m}$, $\tau_{\max} = \frac{T}{W_p} = 19.9 \text{ MPa}$ (2 分)

最大弯矩在固定端: $M_{\max} = F_2 L = 7.5 \text{ kN} \cdot \text{m}$ (2 分)

$\sigma_{\max} = \frac{|F_N|}{A} + \frac{|M_{\max}|}{W_z} = 8.0 + 149.3 = 157.3 \text{ MPa}$ (3 分)

$\sigma_3 = \sqrt{\sigma_{\max}^2 + 4\tau_{\max}^2} = 162.3 \text{ MPa} > [\sigma]$ (3 分)

该悬臂梁不安全 (1 分)